SmartWatch Technical Reference

HRV Formulas · Gyroscope Tilt · Drowsiness Detection Algorithm · Margin of Error

This document provides the mathematical foundations, signal processing formulas, detection algorithm, and expected accuracy margins for the physiological monitoring system implemented on the ESP32-S3 smartwatch platform.

1. Heart Rate Variability (HRV) — Formulas

Heart Rate Variability (HRV) measures the variation in time between consecutive heartbeats, called **RR intervals** (or NN intervals when referring to normal-to-normal beats). A higher HRV generally indicates a healthy, responsive autonomic nervous system. Drowsiness and fatigue suppress HRV.

1.1 RR Interval

The RR interval is the time (in milliseconds) between two successive R-peaks in the ECG/PPG signal — i.e., two consecutive heartbeats.

```
RR[i] = t[i] − t[i−1] (milliseconds)
```

where t[i] is the timestamp of the i-th detected beat peak.

1.2 Mean RR (Average Interval)

```
RR_mean = (1/N) × Σ RR[i] for i = 1 to N
```

N = number of RR intervals in the analysis window (we use N = 20).

1.3 SDNN — Standard Deviation of NN Intervals

SDNN is the primary HRV metric implemented in the system. It reflects overall autonomic variability. Normal awake adults: 50–100 ms. Drowsy/fatigued: typically < 30 ms.

```
SDNN = sqrt[ (1/N) × Σ (RR[i] − RR_mean)² ]
```

Expanded step-by-step:

```
1. Collect last N RR intervals: RR[1], RR[2], ..., RR[N] 2. mean = (RR[1] + RR[2] + ... + RR[N]) / N 3. For each i: diff[i] = RR[i] - mean 4. variance = (diff[1]^2 + diff[2]^2 + ... + diff[N]^2) / N 5. SDNN = sqrt(variance) [result in milliseconds]
```

1.4 RMSSD — Root Mean Square of Successive Differences

RMSSD reflects short-term parasympathetic activity. Not currently displayed but can be added as a secondary metric.

```
RMSSD = sqrt[ (1/(N−1)) × Σ (RR[i+1] − RR[i])² ] for i = 1 to N−1
```

1.5 SDNN Clinical Reference Ranges

SDNN Range (ms)	Interpretation	Drowsiness Risk
> 100 ms	Excellent HRV — fully alert, healthy ANS	Very Low
50 – 100 ms	Normal awake adult range	Low

30 – 50 ms	Mild fatigue or reduced vigilance	Moderate
< 30 ms	Significant HRV suppression — fatigue/drowsiness	High
< 15 ms	Severe suppression — likely drowsy or sensor error	Very High

Table 1. SDNN clinical reference ranges for drowsiness assessment.

2. Gyroscope / Accelerometer Tilt Formula

Tilt angle is computed from the QMI8658C accelerometer axes, not the gyroscope. The accelerometer measures gravitational pull along each axis; when the wrist tilts (as in a drowsy head-drop gesture), the Z-axis component changes relative to total acceleration magnitude.

2.1 Total Acceleration Magnitude

$$|a| = \mathrm{sqrt}(\ ax^2 + ay^2 + az^2\ )\ \text{[in g-units]}$$

where ax, ay, az are the accelerometer readings on each axis in g. When stationary, $|a| \approx 1.0$ g (Earth's gravity).

2.2 Tilt Angle from Vertical

The angle θ between the device's Z-axis and vertical (gravity vector) is computed using the inverse cosine of the normalized Z component:

$$\theta = \arccos(\ az\ /\ |a|\ )\ \text{[result in radians, convert to degrees]}$$

$$\theta\_deg = \theta \times (180\ /\ \pi)$$

In C++ (Arduino) implementation:

```
float mag = sqrt(ax*ax + ay*ay + az*az); float tiltAngle = degrees( acos( constrain(az /
mag, -1.0f, 1.0f) ) );
```

Note: *constrain() clamps the argument of arccos to [–1, +1] to prevent NaN from floating-point rounding errors when az ≈ |a|.*

2.3 Tilt Angle Interpretation

θ (degrees)	Physical Meaning	Drowsiness Signal
0° – 10°	Device level / wrist flat upright	None
10° – 20°	Slight wrist tilt — normal movement	Baseline variation
20° – 35°	Noticeable wrist drop	Mild (monitor)
35° – 55°	Significant forward head/wrist drop	Moderate alert
> 55°	Severe tilt — possible nodding off	Strong alert

Table 2. Tilt angle interpretation for wrist-worn device.

2.4 Tilt Deviation from Baseline

Because individuals naturally hold their wrist at different resting angles, absolute tilt is less meaningful than deviation from the calibrated baseline:

$$\Delta Tilt = |\ \theta\_current - \theta\_baseline\ |\ \text{[degrees]}$$

θ_baseline is measured and averaged over the 3-minute calibration window. ΔTilt > 15° triggers a partial drowsiness score contribution.

3. Drowsiness Detection Algorithm

The system uses a weighted multi-modal scoring algorithm combining three physiological signals. A composite drowsiness score D (0–100) is computed every 500 ms after the calibration phase completes.

3.1 Calibration Phase (0 – 3 minutes)

Before drowsiness detection begins, the system runs a 3-minute (180-second) calibration window to establish the user's personal baseline. During this window:

```
For each detected heartbeat during calibration: — Record RR interval — Compute running SDNN — Record current tilt angle At end of 180 seconds: baselineSDNN = average of all SDNN samples collected baselineTilt = average of all tilt angle samples collected
```

Note: *Calibration must be performed while the user is alert and in a normal seated/standing posture. Results are invalid if calibrated while already fatigued.*

3.2 Composite Drowsiness Score Formula

Three sub-scores are computed and summed:

Sub-score 1 — HRV Drop Score (0 to 50 points):

```
S_hrv = max( 0, ((baselineSDNN − SDNN_current) / baselineSDNN) × 50 )
```

S_hrv = 0 when SDNN ≥ baseline (alert). S_hrv = 50 when SDNN = 0 (maximum suppression).

Sub-score 2 — Tilt Deviation Score (0 to 30 points):

```
S_tilt = min( 30, (ΔTilt / 15°) × 30 )
```

S_tilt = 0 when ΔTilt = 0°. S_tilt = 30 when ΔTilt ≥ 15°.

Sub-score 3 — Low Heart Rate Score (0 to 20 points):

```
S_bpm = min( 20, ((65 − BPM_current) / 20) × 20 ) if BPM < 65
S_bpm = 0 if BPM ≥ 65
```

Drowsiness tends to lower heart rate slightly. Threshold is 65 BPM; maximum contribution at 45 BPM or below.

Total Composite Score:

```
D = clamp( S_hrv + S_tilt + S_bpm, 0, 100 )
```

3.3 Decision Thresholds

Score D	State	Display Color	Action
0 – 39	AWAKE	Green	No action
40 – 69	ALERT	Amber	Caution — monitor closely
70 – 100	DROWSY!	Red	Alert caregiver / vibrate

Table 3. Drowsiness score decision thresholds.

3.4 Algorithm Flowchart (Text Form)

```
START | ■■ [Calibration phase — 180s] | Collect SDNN samples → compute baselineSDNN | Collect tilt samples → compute baselineTilt | ■■ [Detection phase — every 500ms] Read current BPM, SDNN, tiltAngle Compute S_hrv = max(0, ((baselineSDNN - SDNN) / baselineSDNN) * 50) Compute S_tilt = min(30, (|tiltAngle - baselineTilt| / 15) * 30) Compute S_bpm = (BPM < 65) ? min(20, ((65-BPM)/20)*20) : 0 D = clamp(S_hrv + S_tilt + S_bpm, 0, 100) IF D < 40 → AWAKE (green) IF D < 70 → ALERT (amber) IF D >= 70 → DROWSY (red) → trigger alert END LOOP
```

4. Margin of Error — Sensors and Derived Metrics

All sensor readings carry inherent measurement uncertainty. The values below reflect expected accuracy for the specific hardware used in this system (MAX30102 pulse oximeter, QMI8658C IMU, analog PPG sensor on GPIO0).

4.1 Heart Rate (BPM) Accuracy

Source	Error / Uncertainty
Analog PPG (GPIO0) — raw ADC noise	±3–5 BPM at rest
After median + IIR filter	±2–3 BPM at rest
Motion artifact (wrist movement)	±5–15 BPM during movement
Adaptive threshold drift	±1–2 BPM
Overall expected accuracy (still, correct placement)	±2–3 BPM
Clinical-grade reference (MAX30102)	±1 BPM (per datasheet)

Table 4. Heart rate measurement error sources.

4.2 HRV (SDNN) Accuracy

Source	Error / Uncertainty
Timing resolution (millis() @ 1ms tick)	±1 ms per RR interval
Beat detection jitter (peak threshold)	±5–10 ms per RR interval
Accumulated error over N=20 intervals	±3–8 ms on SDNN value
Motion artifact false beats	±10–20 ms SDNN during movement
Overall SDNN error at rest	±5–10 ms (acceptable for drowsiness)
Minimum detectable SDNN change	~5 ms

Table 5. HRV (SDNN) measurement error sources.

4.3 Gyroscope / Accelerometer (Tilt) Accuracy

Source	Error / Uncertainty
QMI8658C accelerometer noise density	±0.07 mg/√Hz
QMI8658C zero-g offset (factory)	±20 mg typical
Tilt angle error at rest (computed)	±0.5° – 1.5°
Tilt angle error during slow movement	±2° – 5°
Vibration / rapid motion artifact	±5° – 15° transient
Temperature drift (no compensation)	±0.1° per °C change
Overall tilt accuracy (slow movement)	±1° – 3°

Minimum detectable tilt change	~1°

Table 6. QMI8658C tilt angle error sources.

4.4 Drowsiness Score Accuracy

Because the drowsiness score D is derived from three noisy signals, its uncertainty is the propagated combination of all sub-score errors:

$$\sigma\_D \approx \mathrm{sqrt}(\ \sigma\_hrv\_score^2 + \sigma\_tilt\_score^2 + \sigma\_bpm\_score^2\ )$$

In practice, with sensors at rest and correct placement:

Sub-score	Typical Error Contribution
S_hrv (HRV drop, max 50 pts)	±3 – 6 points
S_tilt (Tilt deviation, max 30 pts)	±1 – 4 points
S_bpm (Low BPM, max 20 pts)	±1 – 2 points
Total D (0–100 scale)	**±4 – 8 points overall**

Table 7. Propagated drowsiness score uncertainty.

Note: *A score margin of ±8 points means the system cannot reliably distinguish D=40 from D=48. The 40/70 thresholds include an implicit safety margin to account for this. For clinical applications, validation against polysomnography (PSG) reference is recommended.*